

CCP-CNS2026 · Computer Networks · Problem 6: Intrusion Detection

WEEK 4 STATUS REPORT

Advanced Features & Final Integration

Reporting Period: Week 4 | Date: 10 June 2025

1. Team Overview

Member	Role	Contact
Muhammad Ahmed	Network Architect + Project Manager	f2024266974@umt.edu.pk
Hamna Rizwan	Security Engineer + Testing/QA	f2024266956@umt.edu.pk
Eshal Tanveer	Documentation Lead + Automation Lead	f2024266958@umt.edu.pk

2. Work Completed This Week

- NAT reviewed — confirmed not applicable for Problem 6 (PCAP-only analysis, no live topology); decision documented
- Baseline performance report completed: throughput and latency benchmarks derived from PCAP packet inter-arrival times and payload sizes
- Final GNS3 project v3 marked N/A (Problem 6 requires no GNS3 topology per instructor specification)
- All configurations document updated to reflect tool versions: Wireshark 4.x, tshark CLI, Python 3.x dissection scripts
- Final test matrix completed — all 12 test cases executed and results recorded (pass/fail)
- Attack classification finalized: confirmed SYN Flood / DoS based on high-rate TCP SYN packets with no corresponding ACK
- Week 4 status report drafted and compiled by Muhammad Ahmed

3. Deliverables Status

Deliverable	Owner	Status	Location
NAT Configuration (N/A rationale documented)	Muhammad Ahmed	✓ Done	GitHub & Google Drive
Baseline Performance Report (throughput, latency)	Hamna Rizwan	✓ Done	GitHub & Google Drive
Final GNS3 Project v3 + All Configs (N/A documented)	Muhammad Ahmed	✓ Done	GitHub & Google Drive
Final Test Results (Test Matrix with Pass/Fail)	Hamna Rizwan	✓ Done	GitHub & Google Drive
Week 4 Status Report (this document)	Muhammad Ahmed	✓ Done	GitHub & Google Drive

4. Baseline Performance Report

5.1 Methodology

Since Problem 6 is a PCAP-only forensics task (no live GNS3 topology), performance baselines were derived directly from the captured PCAP file using Wireshark and tshark statistical analysis. Metrics reflect observed traffic conditions during and around the attack window.

5.2 Throughput Baseline

Metric	Pre-Attack (Baseline)	During Attack
Average throughput (Mbps)	~0.8 Mbps	~14.2 Mbps (anomalous)
Packets per second (pps)	~120 pps	~2,400 pps
TCP SYN ratio	~5% of traffic	>92% of traffic

5.3 Latency Baseline

Metric	Pre-Attack (Baseline)	During Attack
Avg inter-arrival time (ms)	~8.3 ms	~0.4 ms (flood)
TCP handshake completion	Normal SYN-SYN/ACK-ACK	SYN only — no ACK
Observation window	First 60s of capture	Remaining capture

6. Individual Contributions

Member	Contribution This Week
Muhammad Ahmed	Documented NAT N/A rationale; confirmed GNS3 N/A for Problem 6; compiled and submitted all Week 4 deliverables to GitHub and LMS; authored Week 4 status report
Hamna Rizwan	Executed all 12 final test cases; recorded pass/fail results in test matrix; conducted tshark statistical analysis for baseline throughput and latency; prepared performance report section
Eshal Tanveer	Updated all configs documentation; reviewed and finalized the test matrix formatting; updated Google Drive folder structure; reviewed Week 4 report draft for accuracy

7. Issues & Blockers

Issue	Severity	Resolution / Plan
NAT & GNS3 not applicable for Problem 6	Low	Formally documented N/A rationale in /docs/; no action needed
No other blockers at this time	—	—